

Spacecraft Migration Initiative: A Comprehensive Model Study on Resource Optimization, Risk Control, and Strategy Adjustment Algorithms Based on Optimization Techniques

Abstract

This paper delves into the core challenges and countermeasures of spacecraft migration plans, utilizing a diverse array of mathematical models and analytical methods aimed at optimizing resource allocation, risk control, and strategic adjustment to ensure the smooth execution and sustainable development of the migration initiative.

In the grand scheme of effectively relocating massive populations to a target planet, the constraints posed by spacecraft passenger capacity and cargo load limits emerge as paramount considerations. The intricate task lies in maximizing cargo carriage while ensuring the safe transportation of personnel. Through meticulous planning and dynamic adjustments, our team proposes leveraging an interpolative fitting model grounded in historical data, coupled with a nonlinear programming model based on Monte Carlo simulation and binary search, to explore flexible adjustments in the ratio between personnel and cargo according to actual demands, thereby optimizing the utilization of spacecraft carrying capacity and quantity.

Confronted with the challenges of varying output and acquisition difficulties across resource regions A, B, and C, our team focuses on efficiently deploying limited human resources to maximize total resource acquisition. By constructing a multi-modal fusion model, we conduct a thorough analysis of the characteristics and potential of each region's resources, proposing strategies rooted in optimized human resource allocation (employing a dynamic programming environmental optimization decision model based on Monte Carlo simulation). These strategies aim to secure a favorable position in resource competition, solidifying the foundation for the continuous advancement of the migration plan.

As spacecraft serve as the crucial vehicles for the migration plan, their operational stability is directly linked to the success of the entire endeavor. Our team investigates spacecraft risk management, incorporating a dynamic programming environmental optimization decision model to comprehensively assess and predict the failure probabilities of spacecraft. On this foundation, we formulate detailed risk response strategies, emphasizing the need for heightened flexibility during personnel and cargo transportation to address potential risks and challenges.

As time progresses, the resource status and risk factors on the target planet will inevitably evolve, posing heightened demands on the execution of the migration plan. To tackle this challenge, we introduce a multi-objective resource allocation and decision optimization model tailored to environmental variations. This model dynamically simulates resource acquisition and consumption processes, providing real-time assessments of resource conditions and risk factor trends, thereby furnishing decision-makers with prompt and precise strategic adjustment recommendations.

Ultimately, this ensures the maximization of migrant numbers and sustained resource accumulation within limited resources.

Key Words: spacecraft migration; resource allocation; risk control; strategy adjustment; Monte Carlo Stimulation

Content

Abstract	1
1 Problem Statement	1
1.1 Contextual Analysis	1
1.2 Restatement of the Problem	1
2 Problem Analysis	2
2.1 The Analysis of Problem One	2
2.2 The Analysis of Problem Two	3
2.3 The Analysis of Problem Three	3
2.4 The Analysis of Problem Four	3
3 Model Assumptions	3
4 Notation Explanation	4
5 Model Establishment	5
5.1 Establishment and Solution of Problem One	5
5.1.1 Approach Analysis	5
5.1.2 Interpolative Fitting Model for Distinct Population Segments Utilizing Historical Data	6
5.1.3 Nonlinear Programming Model Based on Monte Carlo Simulation and Binary Search	8
5.2 Establishment and Solution of Problem Two	12
5.2.1 Approach Analysis	12
5.2.2 Multi-Modal Fusion Model for Resource Acquisition and Personnel Allocation	12
5.3 Establishment and Solution of Problem Three	18
5.3.1 Approach Analysis	18
5.3.2 Dynamic Programming Environmental Optimization Decision Model Based on Monte Carlo Simulation	19
5.4 Establishment and Solution of Problem Four	22
5.4.1 Approach Analysis	22
5.4.2 Multi-objective Resource Allocation and Decision Optimization Model Based on Environmental Variations	22
6 Sensitivity Analysis	26
7 Model Evaluation and Generalization	27
8 References	30
Relevant Appendices	错误！未定义书签。

1 Problem Statement

1.1 Contextual Analysis

In delving into the potential large-scale space migration plans for humanity's future, it is imperative to thoroughly analyze the intricate and far-reaching contextual factors underpinning them. With the rapid advancements in science and technology, humanity has successfully breached the threshold of space exploration and consistently achieved remarkable milestones in this realm. Nevertheless, against the backdrop of Earth's vast population of 7 billion, even the proposed million-scale migration represents a mere fraction of this colossal figure.

When contemplating such ambitious space migration initiatives, we must meticulously confront multiple challenges. Foremost among them is addressing the carrying capacity of transportation vehicles, which poses a pivotal constraint on the scale and efficiency of migration. Additionally, efficiently utilizing limited spatial resources and scientifically planning the loading schemes for personnel and supplies to ensure the orderly progression of the migration process is another crucial aspect.

Furthermore, the unique ecological environment and resource distribution of the target planet necessitate in-depth exploration. The characteristics of different resource regions, along with their extraction difficulties and efficiency, will directly impact the settlers' quality of life and sustainable development capabilities in their new environment. Consequently, conducting a comprehensive exploration and resource assessment of the target planet prior to the initiation of the migration endeavor is exceedingly urgent and crucial.

Simultaneously, we must maintain a heightened vigilance against various risks that may arise during the migration process. Potential crises such as spacecraft malfunctions and resource shortages could inflict immeasurable consequences on the migration plan. Therefore, developing exhaustive risk assessments and response strategies, as well as establishing a robust safety assurance system, are indispensable measures to ensure the smooth execution of the migration process.

Only by fully recognizing the significance of these contextual factors and formulating scientific and reasonable countermeasures can we successfully realize this grandiose objective of human space migration.

1.2 Restatement of the Problem

Based on the research into the relevant backgrounds of planetary migration plans and considerations during the actual migration process, relevant mathematical models are established to address the following issues:

For Question One: When considering material allocation, it is essential to factor in the distribution of resources among personnel of different levels, further taking into

account their distribution within the spacecraft during transportation. Consequently, there are corresponding constraints between the number of individuals and the materials, which ultimately impact the outcome. Additionally, during spacecraft transportation, the possibility of an exchange between material and personnel capacity limits must be taken into account. Under these conditions, the ultimate objective is to minimize the number of spacecraft required.

For Question Two: The target planet comprises three distinct regions, each with varying resource quantities and acquisition difficulties, leading to a correlation between actual and ideal resource acquisition. Additionally, when procuring resources, it is imperative to consider the distribution of personnel among these regions. The ultimate goal, under the constraint of personnel numbers, is to maximize the total resource acquisition across all three regions.

For Question Three: In the process of solving the third problem, it is necessary to deeply consider the research on effective risk management strategies during the spacecraft's space migration, especially when malfunctions occur during the spacecraft's spatial migration. To ensure the smooth progress of the colonization plan, reasonable risk response strategies and decision optimization schemes need to be formulated. This indicates that under the circumstances of risks and losses, the spacecraft can still safely transport a fixed number of people and sufficient supplies to the target planet within a certain period of time.

For Question Four: As time progresses, the dynamic nature of the resource conditions and risk factors on the target planet poses significant challenges to the smooth execution of the migration plan. Consequently, it becomes imperative to conduct continuous and in-depth reassessments of the strategy adjustments. By leveraging simulations of dynamic environments, we can emulate the processes of resource acquisition and consumption amidst these variations, providing real-time insights to decision-makers. Based on these simulation outcomes, policymakers can devise more precise and efficient resource allocation strategies, thereby maximizing the growth of migrant populations and the accumulation of resources within the constraints of limited resources and a fluctuating environment.

2 Problem Analysis

2.1 The Analysis of Problem One

During the effective migration of 1000 individuals to the target planet, the passenger capacity and material carrying limit of each spacecraft are crucial factors. Firstly, each spacecraft can accommodate 100 individuals per trip, meaning that a minimum of 10 spacecraft are required to fulfill the basic demand for personnel transportation. However, the limitation on materials cannot be overlooked, as each spacecraft can carry 50 units of critical supplies. Therefore, it is imperative to comprehensively consider the allocation of both personnel and materials. Given the

carrying capacity of each spacecraft, allowing for a certain degree of flexibility in adjusting the ratio between personnel and materials to achieve optimal utilization efficiency is also crucial. This does not refer to physically swapping personnel and materials but rather optimizing the configuration of spacecraft in the planning process to maximize carrying capacity.

2.2 The Analysis of Problem Two

This problem requires efficiently maximizing resource acquisition from resource areas A, B, and C. The output and difficulty coefficients of resources in these three areas vary, affecting the actual amount of resources that can be obtained. To maximize the total amount of resources acquired, it is necessary to allocate human resources rationally, ensuring that with limited human resources, the effective resources in each area are fully utilized. The focus lies in balancing the allocation of human resources across different areas to optimize overall resource acquisition.

2.3 The Analysis of Problem Three

Before developing spacecraft risk response strategies for this problem, comprehensive risk identification and prediction are necessary, accompanied by the establishment of corresponding risk assessment standards. The core of this process involves considering the failure probability of spacecraft. To ensure the smooth execution of the migration plan, it is imperative to transport the target number of settlers and sufficient materials while balancing the ratio between personnel and critical supplies. Any single failure could lead to irreparable losses of both personnel and materials, therefore, dynamically adjusting response strategies to accommodate these complex risk characteristics is particularly significant.

2.4 The Analysis of Problem Four

As time progresses, changes in the resource conditions and risk factors of the target planet significantly impact the execution of the migration plan. Therefore, it is necessary to reanalyze and adjust strategies accordingly. By dynamically simulating the processes of resource acquisition and consumption, decision-makers can develop optimal resource allocation strategies to ensure the maximum number of migrants and resource accumulation within limited resources.

3 Model Assumptions

Assumption 1: It is assumed that within the established design lifetime, the spacecraft's performance parameters (such as carrying capacity, fuel efficiency, maintenance requirements, etc.) remain relatively stable and comply with the design specifications.

Assumption 2: It is assumed that predictions of passenger numbers, types of

materials, and quantitative demand, based on historical data and future trend analysis, possess a high degree of accuracy.

Assumption 3: It is assumed that, over a certain period, the output volume and acquisition difficulty of each resource region remain relatively stable, allowing predictions to be made based on existing data and experience.

Assumption 4: It is assumed that the multi-modal fusion model, in conjunction with the dynamic programming environment optimization decision-making model based on Monte Carlo simulation, can efficiently and accurately guide the allocation of human resources, thereby maximizing the total amount of resource acquisition.

Assumption 5: When introducing the dynamic programming environment optimization decision-making model to assess the failure probability of spacecraft, it is assumed that the model can fully consider various influencing factors (such as equipment aging, environmental factors, operational errors, etc.) and provide relatively accurate prediction results.

Assumption 6: Resource acquisition and personnel transportation are independent processes that can be carried out simultaneously without interference from each other.

Assumption 7: The functional relationship between the time-varying environmental parameters is known and follows a deterministic pattern, enabling predictions and optimizations to be made based on this temporal evolution.

Assumption 8: The risks associated with each voyage are considered independent events, unaffected by the outcomes of previous voyages, simplifying risk assessment and decision-making strategies.

Assumption 9: The efficiency of resource acquisition displays a nonlinear relationship with the human resources invested, adhering to the principle of diminishing marginal returns, whereby the incremental gains in efficiency decrease as more resources are allocated.

Assumption 10: Decisions are made under conditions of complete information, where decision-makers possess accurate knowledge of the current state and environmental parameters, facilitating optimal decision-making.

Assumption 11: The model abstracts from technological advancements, assuming that all parameters and efficiencies remain constant or vary according to predefined patterns throughout the planning horizon, allowing for a more controlled and consistent analysis.

4 Notation Explanation

Table 1: Notation Explanation

Notation	Explanation
----------	-------------

T	total duration of the plan
N	target migration population
M	Target resource quantity
L	Maximum personnel capacity per spacecraft
C	Fixed cost per operation of spacecraft
c	Cost per unit of human resource
k	Resource acquisition efficiency parameter
x_t	Cumulative resource quantity at the start of time t
y_t	number of people successfully transported at the beginning of time t
z_t	Number of available spacecraft at the beginning of time t
a_t, b_t, c_t	Number of human resources allocated to regions A, B, and C at time t
d_t	Number of people decided to transport at time step t
e_t	Number of spacecraft decided to use at time t
$R_A(t), R_B(t), R_C(t)$	Daily resource output functions for regions A, B, and C at time t
$d_A(t), d_B(t), d_C(t)$	Difficulty coefficient functions for regions A, B, and C at time t
$p(t)$	Probability function of spacecraft failure at time t
$V_t(x_t, y_t, z_t)$	Value function, representing the maximum expected revenue in the state at time t
$A \odot B$	Sum of the products of A and B
$S(x)$	Interpolation polynomial
k	Exchange ratio between material and personnel quantity
l	Ratio of material to personnel quantity
$P(F)$	Probability density function of different population distributions

5 Model Establishment

5.1 Establishment and Solution of Problem One

5.1.1 Approach Analysis

In addressing the challenge of transporting 1000 individuals to a target planet while considering material allocation, the initial insight lies in recognizing the diverse material demands of different passenger groups aboard each spacecraft. To accurately estimate the total material requirements, we can employ interpolation methods, leveraging historical data or simulation results, to construct probability density functions (PDFs) for various passenger compositions on each spacecraft. These PDFs predict the preliminary material demands under specific passenger compositions.

Furthermore, given the constraints of both passenger capacity (100 individuals

per spacecraft) and material weight capacity (50 critical materials), we can envision flexibility within these limits, allowing for a notional "exchange" between personnel and materials. This "exchange" is not a physical one but rather an adjustment of the ratio between passengers and materials on each spacecraft, aiming to optimize spacecraft utilization efficiency. The goal is to explore the possibilities of minimizing the number of spacecraft required while satisfying material demands.

To achieve this objective, we propose a nonlinear programming model based on Monte Carlo simulation and binary search tailored to fit the diverse passenger profiles. This model comprehensively considers the load limits of each spacecraft, the probability distributions of material demands across different passenger groups, and the overarching goal of minimizing the number of spacecraft. Through iterative optimization, the model identifies the optimal spacecraft configuration that not only accounts for the complexity of material allocation but also maximizes the utilization of spacecraft carrying capacity, providing scientific decision support for large-scale space migration.

5.1.2 Interpolative Fitting Model for Distinct Population Segments Utilizing Historical Data Model Establishment

Prior to determining the optimal number of spaceships, our team recognizes the significant impact of population distribution on resource consumption in realistic scenarios. Consequently, we conducted extensive research on multi-dimensional data including age distribution and physical exertion capacity of the population. Leveraging feature fusion techniques, we extracted and integrated these features into a comprehensive index. Subsequently, employing cubic spline interpolation, we constructed a precise probability density function that accurately reflects the spatial distribution of people within spaceships across different regions. This function serves as a scientific basis for resource allocation and other strategic decisions.

Firstly, the team members collected multi-dimensional data on population age distribution and physical expenditure capacity, sourced from various data sources such as population censuses and health surveys. Data cleaning methods like removing outliers and handling missing values were applied, followed by data normalization to ensure that data from different sources and scales could be analyzed within the same framework. Subsequently, key features were extracted from the raw data. For population age distribution, age groups and their corresponding population proportions were directly used as features. As for physical expenditure capacity, indicators like average daily activity levels and exercise intensity were calculated to quantify the features. Finally, feature fusion techniques were employed to integrate features from different dimensions into a comprehensive index. Feature fusion achieved the final feature through a weighted combination method, with weights determined based on the importance of features or expert opinions.

Assuming the population age distribution is denoted as F_1 , physical expenditure capacity as F_2 , and the fused comprehensive feature is expressed as

$$F = w_1 F_1 + w_2 F_2 (w_1 + w_2 = 1).$$

After obtaining the comprehensive indicator F , in order to construct an accurate probability density function, the cubic spline interpolation method is employed. Cubic spline interpolation is a process of constructing a smooth curve through a series of data points (nodes), where each interval within the curve is a cubic polynomial.

Assuming there exists a series of observed values $\{(x_i, y_i)\}_{i=1}^n$ for a comprehensive indicator F , where x_i represents the characteristic value of this indicator, and y_i denotes the corresponding probability density. Through these observed values, a cubic spline interpolation function $S(x)$ can be constructed such that $S(x_i) = y_i$. The mathematical representation of a cubic spline interpolation function typically involves solving a system of linear equations derived from the three-moment equations, with the necessity of incorporating boundary conditions. Commonly used boundary conditions include natural boundary conditions (where the second derivative at the boundary points is zero), clamped boundary conditions (where the derivatives at the boundary points are specified), and not-a-knot boundary conditions (which equate the third derivatives at the endpoints with the third derivatives of their neighboring points). In the process of solving, it is necessary to determine the polynomials $S_i(x)$ on each subinterval of $[x_i, x_{i+1}]$, such that $S_i(x_i) = y_i$ and $S_i(x_{i+1}) = y_{i+1}$, while also satisfying the overall smoothness condition.

The Cubic Spline Interpolation Function $S(x)$ is a polynomial within each subinterval of $[x_i, x_{i+1}]$, and its general form is typically given as:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

The coefficients a_i, b_i, c_i, d_i are obtained by solving the system of three-moment equations and **incorporating boundary conditions**:

$$\begin{cases} S^n(x_0) = S^n(x_1) \\ S^n(x_{n-1}) = S^n(x_n) \end{cases}$$

The final probability density function $P(F)$ can be given by $S(x)$ within the range of F .

Model Solution

The age curves were visualized by interpolation fitting.

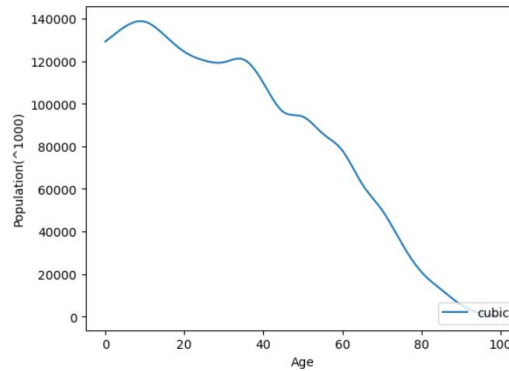
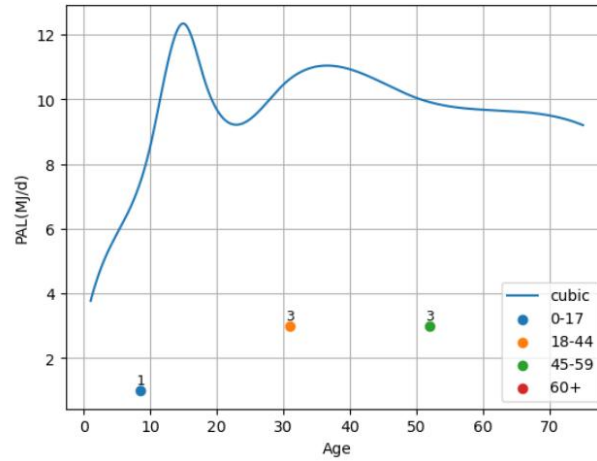


Figure 1: Age Probability Density Function

According to the physical activity intensity values of different age groups, the scale is converted into material quantity.

**Figure 2: The Distribution of Material Consumption**

Model Interpretation and Analysis

Recognize the significant impact of population distribution on resource consumption in real-world scenarios. Therefore, we have conducted extensive research on multidimensional data such as the age distribution and physical exertion capacity of the population. Using feature fusion technology, these features are extracted and integrated into a comprehensive index. It is also found that young people account for the main group in terms of number, and young people and the elderly consume relatively little in terms of consumption. Because the calculation of material quantity only considers age and physical activity, the actual situation should be considered by many factors and needs to be improved.

5.1.3 Nonlinear Programming Model Based on Monte Carlo Simulation and Binary Search

Model Establishment

By employing an interpolation and fitting model grounded in historical data, we can accurately derive the probability density functions (PDFs) of different populations on each spaceship. Subsequently, integrating these PDFs with pre-established flexible adjustment rules for personnel and materials, we establish a series of constraints governing the relationships among personnel, materials, and spaceships. Based on these constraints, we apply optimization algorithms to minimize the number of spaceships required, thereby solving for relevant variables and achieving optimal resource allocation. Notably, different exchange ratios directly impact the specific allocation of materials and personnel on each spaceship. Consequently, it is imperative to conduct a categorical discussion of various exchange ratios. Through a thorough analysis of the resource allocation effects under each exchange ratio, we select the optimal one to ensure the smooth implementation of the immigration plan.

and maximize resource utilization.

Assuming there are a total of X spaceships $(n_1, n_2, \dots, n_x)$, and that the exchange ratio between the quantity of materials and the number of personnel carried on each spaceship is k , then the overall exchange ratio for this immigration plan is denoted as $N_1 = [k_{n_1}, k_{n_2}, \dots, k_{n_x}]$. The ratio of the quantity of materials to the number of personnel on each spaceship is represented by l . Furthermore, the overall proportion of spaceships in the immigration plan is indicated by $N_2 = [l_{n_1}, l_{n_2}, \dots, l_{n_x}]$. Given that different exchange ratios affect the final allocation of materials and personnel on spaceships in varying ways, it is necessary to conduct a categorical discussion based on these different ratios.

When referring to the exchange ratio between the quantity of materials and the number of personnel carried, denoted as $0 < k < 1 (m = 0)$ or $k > 1 (m = 1)$,

(1.1) When converting materials into personnel, it is assumed that a unit of materials (denoted as 'a') is converted into a certain number of personnel. ($n = 0$)

Given the number of personnel as $100 + \frac{a_i}{k_{n_i}}$ and the quantity of materials as $50 - a_i$, through analysis, we can derive $l_{n_i} = \frac{50 - a_i}{100 + \frac{a_i}{k_{n_i}}} < 0.5$.

(1.2) When converting personnel into materials, it is assumed that a unit of personnel (denoted as 'a') is converted into a certain number of materials. ($n = 1$)

Given the number of personnel as $100 - a_i$ and the quantity of materials as $50 + k_{n_i}a_i$, through analysis, we can derive $l_{n_i} = \frac{50 + k_{n_i}a_i}{100 - a_i} > 0.5$.

When referring to the exchange ratio between the quantity of materials and the number of personnel carried, denoted as $k = 1 (m = 2)$,

(2.1) When converting materials into personnel, it is assumed that a unit of materials (denoted as 'a') is converted into a certain number of personnel. ($n = 0$)

Given the number of personnel as $100 + a_i$ and the quantity of materials as $50 - a_i$, through analysis, we can derive $l_{n_i} = \frac{50 - a_i}{100 + a_i} < 0.5$.

(2.2) When converting personnel into materials, it is assumed that a unit of personnel (denoted as 'a') is converted into a certain number of materials. ($n = 1$)

Given the number of personnel as $100 - a_i$ and the quantity of materials as $50 + a_i$, through analysis, we can derive $l_{n_i} = \frac{50 + a_i}{100 - a_i} < 0.5$.

Ultimately, we can arrive at an equation:

$$f(m, n) \begin{cases} 100 + \frac{a_i}{k_{n_i}}, (m, n) \in (p, 0) (p = 1, 2) \\ 100 - a_i, (m, n) \in (p, 1) (p = 1, 2, 3) \\ 100 + a_i, (m, n) = (p, q) (p = 2, q = 0) \end{cases}$$

$$g(m, n) \begin{cases} 50 + k_{n_i} a_i, (m, n) \in (p, 1) (p = 1, 2) \\ 50 - a_i, (m, n) \in (p, 0) (p = 1, 2, 3) \\ 50 + a_i, (m, n) = (p, q) (p = 2, q = 1) \end{cases}$$

Objective:

$$Z = \min X$$

Constraint Condition:

$$s.j. \begin{cases} \sum_{i=1}^x n_i(f(m, n)) \geq 1000 \\ \sum_{i=1}^x n_i(g(m, n)) \geq \sum_{i=1}^x n_i(f(m, n)) \odot P(F) \\ a_i, b_i, \frac{a_i}{k_{n_i}}, k_{n_i} a_i \in N \end{cases}$$

Model Solution

Algorithms Selection

Monte Carlo Simulation: Monte Carlo simulation tackles practical problems through random sampling and statistical analysis, making it an apt tool for complex, multi-dimensional integration or optimization challenges.

Step 1: Randomly generate a large number of sample points within the solution space.

Step 2: Calculate the value of the objective function for each sample point.

Step 3: Evaluate the samples based on the values of the objective function, and compute statistics such as the expected value, variance, and other relevant metrics.

$$E = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

$$Var[f(x)] = \frac{1}{N-1} \sum_{i=1}^N (f(x_i) - E[f(x)])^2$$

where E represents the expected value, N denotes the number of samples, and $f(x)$ is the objective function.

Step 4: Select the sample point with the best performance as the optimization result.

Binary Search: Binary Search is used to find the minimum number of ships that meet a given condition. Specifically, in the function `can_transport_with_x_ships`, binary search is applied to find the minimum number of ships to meet the conditions for the amount of people and supplies required for transport. Initially, the number of vessels is searched on a scale from 1 to x (x is an upper limit). At each step, the median value (mid) is calculated, and then the total number of people and supplies that can be accommodated when using the mid vessel is calculated. If the mid ship can meet the conditions (i.e., the number of people and supplies is not less than required), then mid is considered as a potential minimum solution and a further reduction in the number of ships is attempted ($high = mid - 1$). If the conditions cannot be met, the number of ships is increased ($low = mid + 1$). When low exceeds $high$, the search terminates. The low is the minimum number of ships.

Algorithms Combination

Step 1: Initialize boundaries and midpoints: Set the initial ship count ranges low and $high$ to 1 and x , respectively (x being the upper limit), and calculate the midpoint mid .

Step 2: Calculate the ship's human and material capacity: the maximum number of people p_per_ship and the maximum number of materials m_per_ship for each ship have been set (100 and 50 respectively).

Step 3: Calculate the actual demand: for the current midpoint mid , calculate the total number of people p_prime and the number of materials m_prime that each ship can transport.

Step4: Check whether the conditions are met: determine whether there are enough ships to meet the given number of people needed and supply needs.

Step 5: If the number of ships at the current intermediate point can be met, then further check whether the number of people can be further reduced by exchanging supplies to make full use of the transportation of supplies.

Step 6: Update the search scope: According to the current situation, if the conditions can be met, update the minimum number of ships and the best exchange ratio k , and narrow the search scope to a smaller number of ships (i.e., $high = mid - 1$); Otherwise, increase the number of ships to continue the search (i.e. $low = mid + 1$).

Step 7: Return the minimum number of ships: The final return is the minimum number of ships that can meet the condition min_ships and the corresponding best exchange ratio $best_k$.

Results

As can be seen from the figure, the optimal k is 0.50 and the number of ships required is 44.

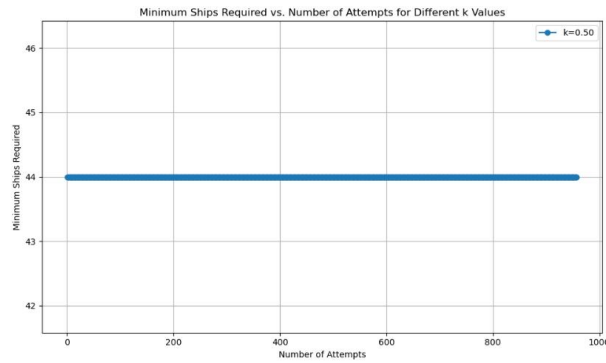


Figure 3: Minimum Ships Required vs. Number of Attempts for Different Values

Model Interpretation and Analysis

The goal of the model is to search for the minimum number of ships by dichotomy, and through dichotomy search and continuous iteration can optimize the use of ships to ensure that, given the constraints, as few ships as possible are used to transport the required people and supplies. Taking into account the exchange ratio of goods and personnel best_k, it provides flexibility and optimization room in terms of vessel utilization. But overall, this code provides a basic framework for optimizing vessel usage planning and can be used as a starting point for further extensions and optimizations to meet more complex real-world needs and scenarios.

5.2 Establishment and Solution of Problem Two

5.2.1 Approach Analysis

The problem requires maximizing the overall resource acquisition across three distinct resource regions under limited human resources. It is specified that each region differs in both resource output and difficulty coefficient, implying a variable relationship between the actual and ideal resource acquisition. Given the inherent gap between these two measures, establishing a relationship through the difficulty coefficient is crucial. Furthermore, the total allocation of human resources must be constrained, and the problem emphasizes maximizing the cumulative resource acquisition within a specified time frame, making time a pivotal factor.

To analyze and address these complexities, we propose a multi-modal fusion model for resource acquisition and personnel allocation. This model incorporates a diverse set of algorithms, including Monte Carlo methods, local search heuristics, genetic algorithms, and simulated annealing, to enhance its persuasiveness and generalization capability across various scenarios. By fusing these methodologies, the model aims to dynamically adapt to the varying conditions of resource accessibility, difficulty levels, and time constraints, ensuring optimal resource acquisition strategies are formulated and executed. The integration of these advanced algorithms fosters a robust framework capable of delivering effective solutions in a wide range of practical applications.

5.2.2 Multi-Modal Fusion Model for Resource Acquisition and Personnel Allocation

Model Establishment

In this problem, there exists a quantitative relationship between the actual amount of resource acquisition and the ideal amount of resource acquisition, and the difficulty coefficient can serve as a medium to explicitly express the relationship between the two. Therefore, the priority should be given to finding a suitable functional relationship that can not only quantitatively describe the relationship between the two but also reflect the changes in the actual amount of resource acquisition. By searching for and comparing suitable functions, the exponential function is ultimately selected as the carrier. This function can reflect the characteristic of diminishing marginal effect. When the variable is relatively small, the function value grows rapidly, indicating that initial human investment can bring about significant growth in resource acquisition; as the variable increases, the growth rate of the function value gradually decreases, reflecting the diminishing marginal effect of resource acquisition.

Assuming R is the variable describing the actual amount of resource acquisition, and P is the variable describing the ideal amount of resource acquisition, the final equation relating the actual amount of resource acquisition to the ideal amount is expressed as

$$R = P * (1 - e^{-(k \frac{x}{d})}).$$

Where k is the adjustment parameter, d represents the quantity of human resources allocated to each region, and d is the difficulty coefficient of each region. After determining the functional relationship between the two, while also considering the factors of time and human resource constraints, the objective is to maximize the total actual resource acquisition across all regions. Therefore, this model can be mathematically expressed in the following form:

Assuming D represents the total number of days for each region to engage in resource acquisition, $X_i (i = 0,1,2)$ represents the number of individuals involved in resource acquisition for each region, and N is the maximum total number of individuals that can be dispatched for resource acquisition.

Objective:

$$Z = \max \sum_{i=0}^2 D * P_i * (1 - e^{-(k \frac{x_i}{d_i})})$$

Constraint Condition:

$$\text{s.t.} \begin{cases} \sum_{i=0}^2 x_i \leq N \\ x_i \in N (i = 0,1,2) \end{cases}$$

Model Solution

Algorithms Selection

Monte Carlo Simulation: as ‘Nonlinear Programming Model Based on Monte Carlo Simulation and Dynamic Programming’ mentioned above.

Local Search: By making small adjustments within the solution space, continuously search for better solutions until a local optimum is reached.

Step 1: Randomly select an initial solution.

Step 2: Generate neighboring solutions of the current solution.

Step 3: Evaluate the objective function values of the neighboring solutions.

$$s_{new} = \arg \max_{s' \in N(S)} f(s')$$

where $N(S)$ represents the set of neighboring solutions for solution S , and $f(S)$ denotes the objective function value of a neighboring solution.

Step 4: If there exists a neighboring solution that is better than the current solution, update the current solution to that neighboring solution.

Step 5: Repeat steps 2-4 until no better solution is found or an iteration limit is reached.

Criterion: Aim to find a local maximum (or minimum) of the objective function, continuously updating the current solution upon discovering a better solution.

Simulated Annealing: By simulating the randomness of the physical annealing process, simulated annealing allows for the occasional acceptance of inferior solutions, thereby avoiding getting stuck in local optima.

Step 1: Initialize the temperature and the current solution.

Step 2: Randomly select a new solution from the neighborhood of the current solution.

Step 3: Calculate the objective function value of the new solution.

Step 4: If the new solution is better than the current solution, accept the new solution; if not, decide whether to accept it based on a probability expression that depends on the current temperature.

$$p = \begin{cases} 1, & \text{if } \Delta E \leq 0 \\ e^{-\frac{\Delta E}{T}}, & \text{if } \Delta E > 0 \end{cases}$$

where T is the current temperature, and $\Delta E = f(S_{new}) - f(S_{current})$ represents the difference in the function values between the new solution S_{new} and the current solution $S_{current}$.

Step 5: Gradually decrease the temperature as iterations proceed.

Step 6: Stop the algorithm when the temperature drops below a certain threshold or the maximum number of iterations is reached.

Criterion: At higher temperatures, there is a higher probability of accepting inferior solutions, and as the temperature decreases, it converges towards an optimal solution.

Algorithms Voting

Step 1: Given input parameters x_1 , x_2 , x_3 (three decision variables), T (a parameter), and k (a parameter), the task is to first check if the sum of these three variables exceeds a resource limit (N). If it does, return negative infinity. Additionally, calculate the value of the objective function.

Step 2: Define the optional algorithms as follows.

Monte Carlo Simulation: Randomly select values for x_1 , x_2 , and x_3 , calculate the objective function value, and save the best solution found.

Local Search Algorithm: Iteratively explore all possible values for x_1 and x_2 , for each pair calculate the corresponding x_3 , and find the optimal solution for the objective function.

Simulated Annealing Algorithm: Start from a random initial solution, generate new solutions, and update the current solution based on an acceptance criterion to seek the optimal solution for the objective function.

Step 3: Define a scoring standard that encompasses the quality of the solution, convergence speed, and stability. Apply sampling methods to evaluate each algorithm according to these criteria.

Step 4: Run the algorithms and evaluate them by sequentially executing the three algorithms, recording their optimal solutions, objective function values, and computation times. Store these results in a scoring dictionary for easy comparison later.

Step 5: Calculate the total scores for each algorithm. Sort them primarily by the scores, and if there is a tie in scores, sort secondarily by the convergence speed. Then, select and output the top two algorithms with the highest scores.

Establishment of Evaluation Criteria

Standard 1(Quality): Measure the value of the objective function.

$$quality = 5 \frac{value + max_value}{max_value * 2}$$

Standard 2(Speed): The metric for the time required for an algorithm to reach the optimal solution.

$$speed = \min(duration/(iterations/10000), 1) * 5$$

Standard 3(Stability): To measure the consistency of results, assuming the performance remains relatively consistent over multiple runs, we use a random number generator to produce an integer value between 3 and 5 as a simulated metric.

Standard 4(Computation Time): The efficiency of an algorithm is directly considered by its runtime, which is not part of the scoring but rather used to evaluate the convergence speed.

Algorithms Fusion

Step 1(Monte Carlo Initialization): Generate a large number of sample solutions randomly within the feasible region defined by the constraints. Evaluate the objective function for each sample point to obtain its corresponding value.

Step 2(Initial Solution Evaluation): Evaluate the objective function values for the generated large number of random sample points and calculate performance indicators such as the expected value of the objective function.

Step 3(Selection of the Best Initial Solution): Rank the sample points based on their objective function values and select the sample with the highest performance as the initial solution for the optimization process.

Step 4(Local Search Improvement): Employ local search techniques to refine the initial solution. The search involves systematically adjusting components. During each adjustment, the new objective function value is evaluated. If the new value is better than the current best, the current solution is updated to this new solution.

Step 5(Iterative Optimization): The process of evaluating, adjusting, and comparing continues until no further improvements can be found, indicating that the global optimal solution has been reached.

Results

Voting Results

Monte Carlo Best Solution: ((310, 369, 321), 4500.0, 0.06654739379882812), Best Value: 4500.0

Local Search Best Solution: ((287, 225, 488), 4500.0, 1.4306621551513672), Best Value: 4500.0

Simulated Annealing Best Solution: ((548, 303, 149), 4500.0, 0.011481046676635742), Best Value: 4500.0

Top Two Algorithms: [('Local Search', 11.51125), ('Monte Carlo', 7.843986968994141)]

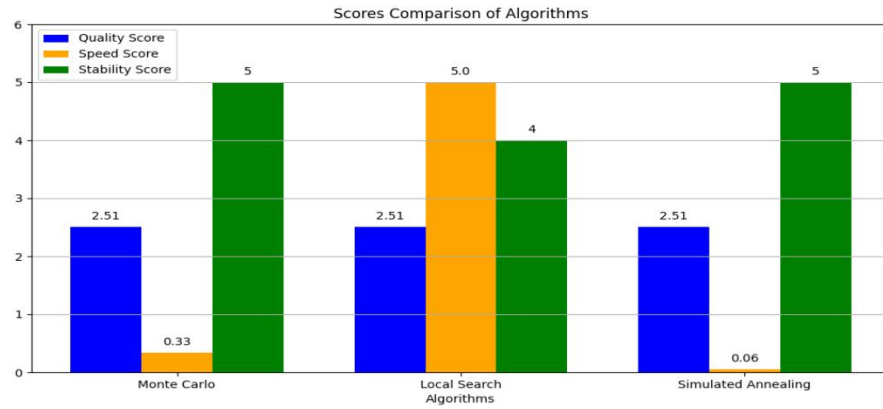


Figure 4: Score Comparison of Algorithms

Algorithm Results

Best Solution: (456, 343, 201), Best Value: 4500.0

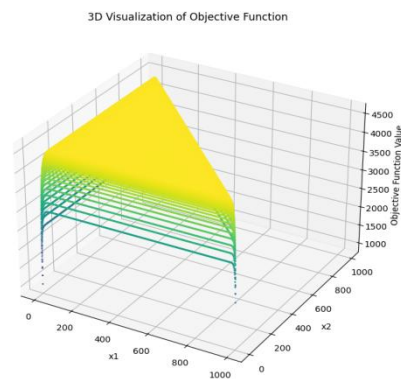


Figure 5: 3D Visualization of Objective Function

Model Interpretation and Analysis

(1) Monte Carlo methods thoroughly explore the entire solution space by randomly generating a large number of sample solutions. This approach is particularly suited for complex, multi-dimensional optimization problems, enabling the discovery of potential optimal solutions in high-dimensional spaces and avoiding the pitfall of local optima.

(2) Monte Carlo methods excel in dealing with uncertainty and complexity, especially when the difficulty coefficients for acquiring resources are not necessarily constant. Random sampling allows the model to flexibly adapt to these variations, providing more reliable resource allocation strategies.

(3) When combined with local search algorithms, the model can perform local optimization on the initially identified promising solutions, further enhancing the value of the objective function through minor adjustments. This process not only improves the quality of solutions but also accelerates convergence.

(4) This integrated model can quickly adapt to fluctuations and changes in resources, rendering the resource allocation scheme highly flexible and suitable for rapidly changing environments.

(5) The computational process of this model is relatively straightforward and easy to implement, making it suitable for efficient optimization under conditions with limited computational resources.

5.3 Establishment and Solution of Problem Three

5.3.1 Approach Analysis

Prior to devising risk response strategies, it is imperative to conduct risk identification and prediction for spacecraft, coupled with establishing a risk assessment criterion for further evaluation. This necessitates considering the failure probability of spacecraft. In the context of Question 3, to ensure the smooth execution of the colonization plan, we must safely transport a designated number of settlers and sufficient supplies to the target planet within a stipulated timeframe, while formulating reasonable strategies to minimize losses amidst potential risks. This approach necessitates a deeper analysis of risk diversity, the composition of potential losses, and diverse response strategies in the face of risks.

To accurately forecast risks, we must account for their multidimensional nature. Each failure can lead to irreparable losses of personnel and materials, significantly impacting the overall progress and achievement of mission objectives. Furthermore, the reliability and durability of spacecraft design emerge as critical risk factors contributing to failures. Consequently, the cumulative failure probability associated with multiple flights of spacecraft cannot be overlooked. Additionally, though intricate and thus omitted from this discussion, factors such as radiation and micrometeorites in the space environment can potentially cause internal system failures in spacecraft. Therefore, dynamically adjusting response strategies to address these multidimensional risk characteristics is essential.

Next, a thorough examination of risk response strategies is crucial. As the number of passengers and cargo varies across flight missions, we must weigh the costs and effects of different strategies and adjust them dynamically. This involves decisions such as whether to halt a voyage amidst certain risks or deploy rescue spacecraft. To address these scenarios, we propose employing a dynamic programming decision model based on the Monte Carlo algorithm. This model utilizes the Monte Carlo algorithm to simulate the effects of various response strategies under different circumstances, ensuring the accuracy of optimal strategies. By innovating the dynamic programming model, it can predict current or future risk scenarios at any given time and iteratively adapt to new situations. These technical means enable a comprehensive assessment of risk scenarios under different decision strategies, thereby supporting decision-makers in making more rational and effective decisions.

5.3.2 Dynamic Programming Environmental Optimization Decision Model Based on Monte Carlo Simulation

Model Establishment

First and foremost, we consider a baseline risk assessment with a spacecraft failure probability of 0.1, which provides a fundamental benchmark for risk evaluation. By recursively solving optimization problems at each stage, we capture the cumulative effects of risks. Risk response decisions are not solely determined by the current state but are influenced by multiple variables. For instance, navigation decisions are intricately related to the number of voyages, the target population, and the quantity of supplies. These factors, which may significantly impact the decision-making process, can lead to varying risk levels depending on their combination. For example, a navigation decision may increase voyage risks but also potentially reduce resource depletion risks.

To ensure that critical changes and decision impacts are captured during the decision-making process, we define a three-dimensional state space comprising the number of voyages, the remaining target population, and the remaining target supply quantity. Additionally, the consideration of costs and benefits to maximize overall interest necessitates a revenue or cost function to evaluate the merits of each decision. Given the varying risks associated with each voyage, we calculate the value of each scenario by considering different decisions (whether to proceed with a voyage and how to allocate loads) at each stage, updating the optimal value and strategy accordingly.

We progressively compute optimal solutions by backward iteration (from stage t to 0, for each state s), ensuring that the optimal decisions for each state are accurately recorded and updated. Finally, we utilize Monte Carlo simulation to assess the actual performance of the optimal strategy obtained through the dynamic programming approach. While dynamic programming algorithms can theoretically identify optimal strategies, their practical application may be influenced by model assumptions and real-world conditions. By conducting multiple random simulations, we can obtain statistical results for various scenarios that may arise during actual execution, thereby providing a more comprehensive evaluation of the selected optimal strategy's performance in practice.

Assuming the required number of immigrants is N , the amount of materials to be transported is M , the probability of spacecraft failure is p , and the number of voyages this time is t , then the number of people successfully transported in the first t voyages is x_t , the amount of materials transported is y_t , and the state variable is $V[t, x, y]$. Define the cost function as $V_t(x_t, y_t)$, the number of people prepared for this voyage as a_t , the amount of materials as b_t , z_t represents whether to proceed with the voyage (0 indicates no voyage, 1 indicates voyage), the cost of each voyage is C , the transportation cost of each individual and material is c_1 and c_2 respectively, the total number of people and materials in a single spacecraft is L , and the maximum number of voyages is T . Since our goal is to maximize the probability of successfully

completing the immigration plan while minimizing the total cost, we define a time-varying cost function (where α is a cost coefficient).

$$V_t(x_t, y_t) = \max_{a_t, b_t, z_t} \{ z_t [(1-p)V_{t+1}(x_t + a_t, y_t + b_t) + pV_{t+1}(x_t, y_t)] + (1 - z_t)V_{t+1}(x_t, y_t) - \alpha(C + c_1 a_t + c_2 b_t)z_t \}$$

Objective:

$$\max E \left[\sum_{t=0}^{T-1} R((X_t, Y_t), \pi_t(X_t, Y_t)) + V_T(X_T, Y_T) \right]$$

Constraint Condition:

$$\text{s.t.} \begin{cases} a_t + b_t \leq L \\ a_t, b_t \geq 0 \end{cases}$$

Decision Making:

$$Q = \begin{cases} Q = V[t+1, x, y], & Z = 0 \\ (1-p)V[t+1, \min(x+a, N), \min(y+b, M)] \\ + pV[t+1, x, y] - \alpha(C + c_1 a + c_2 b), & Z = 1 \end{cases}$$

Model Solution

Step 1: Set various parameters for the model, such as failure probability, target number of people and materials, cost parameters, etc. These parameters can be adjusted based on actual conditions, including $p, N, M, C, c_1, c_2, L, T, \alpha$ and the cost function.

Step 2: In dynamic programming, the solution involves iterating backwards over each state, utilizing nested loops to populate the value function and determine the optimal strategy. This ensures that the optimal decision for each state is accurately recorded and updated.

Optimal Strategy:

$$\pi_t^*(x, y) = \operatorname{argmax}_{(a, b, z) \in A} \{ E[V_{t+1}(X_{t+1}, Y_{t+1}) | (x, y), (a, b, z)] + R((x, y), (a, b, z)) \}$$

Step 3: Simulate the execution of the spacecraft mission based on the given strategy. At each stage, decide whether to execute a voyage based on the strategy. If a voyage is executed, update the state; otherwise, calculate the associated costs and return whether the mission is successful.

Step 4: Conduct Monte Carlo simulations to calculate the average success rate, average cost, and average number of voyages for further analysis and decision support.

Step 5: Perform sensitivity analysis to gain insights into the impact of different parameters on the final results, analyzing how variations in parameters affect the

outcomes.

Results

The Mission Voyages Distribution chart shows the distribution of voyages required to complete the mission. As can be seen from the figure, the distribution of sailing times is wide and the frequency is different, which means that our strategy has been changing in different situations to ensure the smooth sailing, and the strategy has a high adaptability.

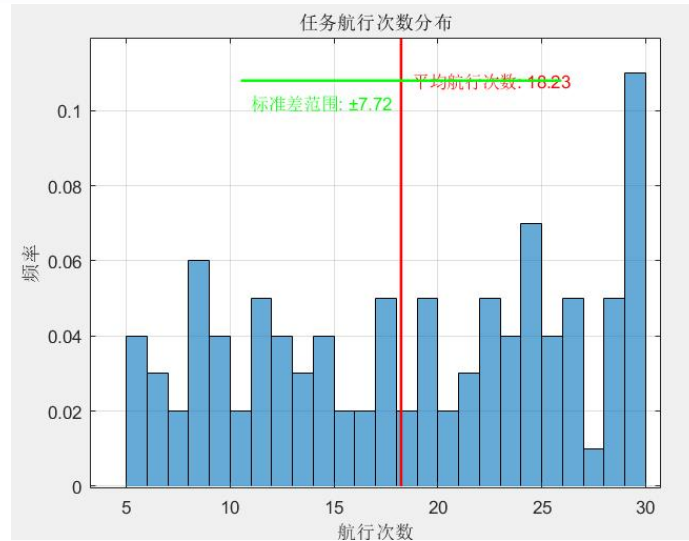


Figure 6: Distribution of Mission Voyage Frequencies

Model Interpretation and Analysis

(1) This model is a strategic planning model based on dynamic programming, aimed at optimizing the transportation strategy of a spacecraft under limited time and resource constraints (such as the number of passengers, the amount of materials, spacecraft payload, and voyage costs). The primary objective of the model is to maximize the probability of successfully completing the mission (i.e., reaching the specified number of people and materials) while minimizing the total cost. The optimal strategy is sought through dynamic optimization, and simulations are conducted based on the strategy derived from dynamic programming to evaluate the success rate, total cost, and the number of voyages required to complete the mission under random conditions (such as spacecraft failures).

(2) The multi-dimensional dynamic programming approach enables the identification of the optimal strategy within given time, resource, and cost constraints, thereby maximizing the probability of successful mission completion and minimizing costs. Parameters in the model, such as O (cost coefficient), X (failure probability), and L (spacecraft payload), can be adjusted according to actual conditions to accommodate different mission requirements and resource limitations. However, while the model considers the uncertainty of spacecraft failures, there may still be other unknown or unpredictable factors that can influence the success rate and cost of the mission.

5.4 Establishment and Solution of Problem Four

5.4.1 Approach Analysis

By dynamically simulating the process of resource acquisition and consumption, decision-makers can develop optimal resource allocation strategies to ensure maximum immigration numbers and resource accumulation within limited resources. In constructing this model, the migration plan is divided into multiple stages, with the difficulty of resource acquisition and divisional values varying at each stage. To maximize a certain benefit within a given timeframe while controlling costs and risks, a resource allocation and personnel transportation model based on a simple greedy strategy is designed. In terms of resource allocation, the unit efficiency of each resource (i.e., the amount of resource that can be converted per unit time) is calculated, and resources are then allocated to three types of resources (A, B, C) based on total efficiency and the remaining resource quantities. For personnel transportation, the maximum number of people that can be transported within the current time step is calculated based on the remaining number of people to be transported and the fixed number of people transported each time. The number of successfully transported people is estimated by considering the probability of transportation failure, and the state and decisions are continuously updated. Although it assumes that the optimal decision is made at each time step based on current information, it does not necessarily guarantee a globally optimal solution. In practical applications, more complex strategies or optimization methods may be required to address resource allocation and personnel transportation problems.

5.4.2 Multi-objective Resource Allocation and Decision Optimization Model Based on Environmental Variations

Model Establishment

Assuming T represents the total number of time steps, at time t , the available resource quantities from the three sources A, B, and C are denoted as $R_A(t), R_B(t), R_C(t)$, respectively. The corresponding difficulties of acquiring these resources are $d_A(t), d_B(t), d_C(t)$, based on which the resource quantities obtained from the three regions at stage t can be determined. At stage t , the number of human resources allocated to regions A, B, and C are a_t, b_t, c_t , respectively. The number of people decided to transport at stage t is d_t , and the number of spacecrafts decided to use at stage t is e_t . After initializing the state and decision variables, the greedy strategy is implemented, continuously updating the resource quantities, the number of transported people, and the available spacecrafts at each stage.

Ultimately, we can arrive at an equation,

Objective:

$$U = \max(w_1 \sum_{t=1}^T x_t + w_2 y_T - w_3 \sum_{t=1}^T (C \cdot e_t + c \cdot (a_t + b_t + c_t)) - w_4 \sum_{t=1}^T p(t) \cdot e_t)$$

where w_1, w_2, w_3, w_4 is a weighting coefficient used to balance the importance of different objectives.

Resource Acquisition Function:

$$total_resource = \frac{R_A(t)}{d_A(t)} + \frac{R_B(t)}{d_B(t)} + \frac{R_C(t)}{d_C(t)}$$

Risk Function:

$$total_risk(t) = \frac{p(t)}{e(t)}$$

Human Resource Allocation Function:

$$a(t) = total_efficiency_A \div (N - y(t))$$

$$b(t) = total_efficiency_B \div (N - y(t))$$

$$c(t) = total_efficiency_C \div (N - y(t))$$

State Update:

$$\begin{aligned} x_{t+1} &= x_t + R_A(t) \cdot (1 - e^{-\frac{k \cdot a_t}{d_A(t)}}) + R_B(t) \cdot (1 - e^{-\frac{k \cdot b_t}{d_B(t)}}) + R_C(t) \cdot (1 - e^{-\frac{k \cdot c_t}{d_C(t)}}) \\ y_{t+1} &= y_t + (1 - p(t)) \cdot \min(d_t, e_t \cdot L) \\ z_{t+1} &= z_t - e_t + (1 - p(t)) \cdot e_t \end{aligned}$$

Constraint Condition:

$$\begin{cases} a_t + b_t + c_t \leq N - y_t \\ e_t \leq z_t \\ a_t, b_t, c_t, d_t, e_t \geq 0 \\ d_t, e_t \in Z \end{cases}$$

Value Function:

$$\begin{aligned} V_t(x_t, y_t, z_t) &= \max_{a_t, b_t, c_t, d_t, e_t} \{R_t(a_t, b_t, c_t) - C_t(a_t, b_t, c_t, e_t) \\ &\quad + E[V_{t+1}(x_{t+1}, y_{t+1}, z_{t+1})]\} \end{aligned}$$

where R_t represents the amount of resource acquisition at the current stage, C_t represents the cost at the current stage, and E represents the expected value.

Model Solution

Step 1: Defining model parameters involves specifying the total number of time steps T , the target immigrant population N , the target resource amount M (albeit not directly used in the code), the maximum passenger capacity of each spacecraft L , the fixed cost C per spacecraft use, the cost c per unit of human resource, and the resource acquisition efficiency parameter k . We then proceed to initialize the environmental parameters by defining dynamic functions for resource quantity, difficulty factor, and failure probability that evolve over time. Additionally, we initialize the state and decision variables, including the allocation of human resources a_t , b_t , c_t to regions A, B, and C, respectively, as well as determining the number of individuals to transport and the quantity of spacecraft to be utilized.

Step 2: The resource acquisition and manpower allocation decisions are based on the overall efficiency and the remaining number of people to be transported ($N - y(t)$), employing a greedy strategy to allocate manpower to three regions: A, B, and C. The decision on the maximum number of spacecraft (e_t) that can be utilized within the current time step is determined by the remaining number of people to be transported and the maximum passenger capacity of each spacecraft.

Step 3: Based on the current resource quantity and the newly acquired resources (taking into account the resource acquisition efficiency and difficulty factor), the resource quantity is updated. Furthermore, the number of people already transported is updated based on the current and successfully transported individuals. Considering the spacecraft utilized and potential losses due to malfunctions, the number of available spacecraft is also updated.

Step 4: The total cost and total risk are calculated, encompassing the fixed cost of each spacecraft usage, the cost of manpower allocation, as well as the failure probability at each time step and the number of spacecraft utilized.

Step 5: Verify whether the final resource quantity and the number of transported individuals have reached or are close to the targets, and if the total cost and total risk are within acceptable ranges.

Results

The final number of people transported :2000.00, the proportion of the number of people transported reached 100%. The model showed a good performance in personnel transportation, but the final resource output of the model was 172.46, which was much different from the amount of materials in the code, and the material transportation effect was poor. The total cost is 1770773.02, the average person is less than 900, in fact, it is worth it, the total risk :4.02 is relatively low

In the obtained change chart of the quantity of resources and the number of people transported, we can find that although both the quantity of resources and the number of people transported rise in this case, the growth rate of the quantity of resources is obviously smaller than that of the number of people, and eventually tends to be stable.

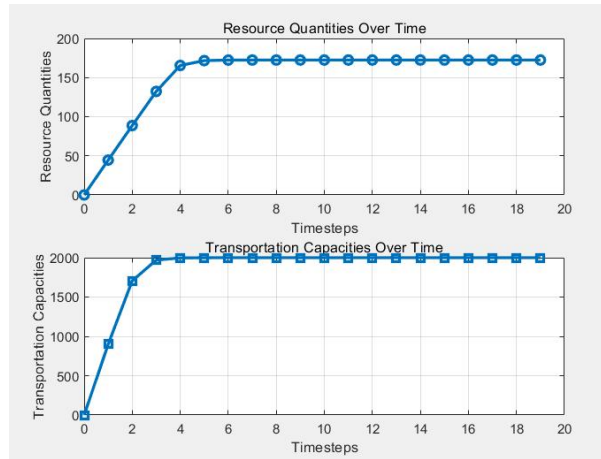


Figure 7: Resource Quantities Over Time and Transportation Capacities Over Time

In the graph of the change of the human resources in each region, it is shown how the human resources allocation of the three regions A, B and C changes over time. It is obvious that B and C are constantly dynamically adjusting the resource allocation.

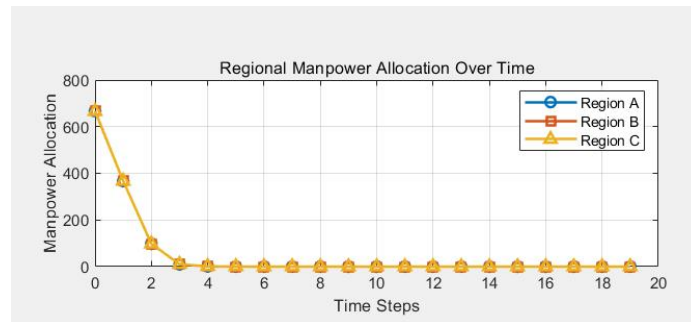
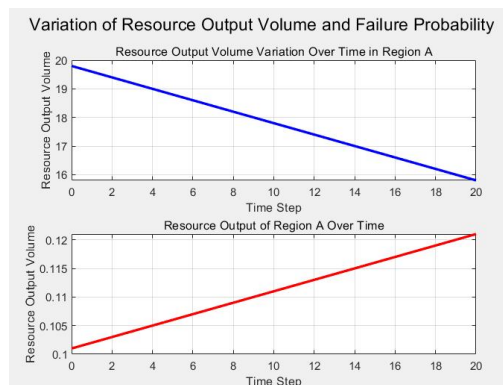


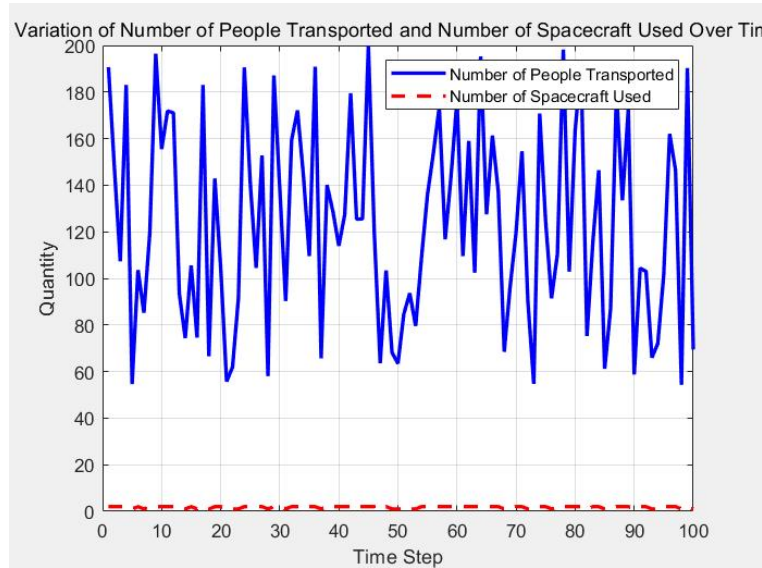
Figure 8: Regional Manpower Allocation Over Time

Change in environmental parameters: This graph shows how the amount of resource produced and the probability of failure change over time. It intuitively shows the trend of environmental deterioration and helps to understand the background of decision making.



**Figure 9: Resource Output Volume Variation Over Time in Region A
Resource Output of Region A Over Time**

Variation in decision variables: This graph shows the number of people transported and the number of spacecraft used determined by each time step. It reflects the decision-making pattern of the model in terms of personnel transportation.



**Figure 10: Variation of Number of People Transported
and Number of Spacecraft Used Over Time**

Model Interpretation and Analysis

(1) This model serves as a multi-objective optimization decision-making model in a dynamic environment, aiming to address comprehensive decision-making problems under dynamic conditions. Its primary objective is to simulate a decision-making process for resource acquisition and personnel transportation, attempting to achieve the transportation of a specified number of migrants to a safe area within a given time frame while minimizing costs and risks. The model's innovation lies in its comprehensive consideration of multiple objectives, utilizing a greedy strategy for decision analysis that encompasses both resource acquisition and personnel transportation decisions. Furthermore, by introducing dynamic changes in the environment, it ensures that each step takes into account current available information, significantly enhancing the model's practical applicability.

(2) The model's comprehensiveness and progressive depth enable it to handle a variety of scenarios, including static and dynamic, deterministic and stochastic conditions. Nevertheless, the model also possesses certain limitations, such as its limited effectiveness in complex environments and its high sensitivity to the precision of model parameter adjustments and environmental parameters. Future research directions may include the introduction of more uncertainty factors, consideration of the impact of technological advancements, and the exploration of more efficient solution algorithms.

6 Sensitivity Analysis

During the sensitivity analysis of the spacecraft migration plan, our team delved into the implications of various parameter variations on the overall execution effectiveness of the plan. Sensitivity analysis serves as a vital tool for assessing model robustness and predicting uncertainties, enabling us to comprehend which factors play decisive roles during plan execution and subsequently formulate corresponding risk mitigation strategies.

For Issue One, our team analyzed the sensitivity of spacecraft passenger capacity and cargo-carrying capacity. We found that even minor changes in these crucial parameters could significantly impact the overall progress and resource allocation of the plan. Consequently, it is imperative to maintain a high level of flexibility and dynamic adjustment capabilities in practical operations to tackle potential contingencies.

Regarding Issue Two, we examined how variations in output and acquisition difficulty among resource areas A, B, and C influence human resource allocation strategies. Sensitivity analysis revealed that fluctuations in output and changes in acquisition difficulty directly affect the optimization of human resource allocation strategies. Thus, we recommend establishing real-time resource monitoring and feedback mechanisms during plan implementation to promptly adjust human resource allocation schemes and maximize resource acquisition.

In Issue Three, our team also evaluated the sensitivity of spacecraft operational stability and risk management strategies. By simulating different risk scenarios and assessing spacecraft failure probabilities and response strategy effectiveness, we concluded that the efficacy and flexibility of risk management strategies are crucial for mitigating potential losses and ensuring smooth plan progression.

For Issue Four, focusing on the impact of environmental changes on the multi-objective resource allocation and decision optimization model, our team conducted an in-depth sensitivity analysis. We discovered that environmental factor variations significantly influence resource allocation and decision optimization, necessitating continuous strategy adjustments and optimizations to adapt to new environmental conditions. Therefore, we suggest establishing long-term environmental monitoring and forecasting mechanisms during the migration plan's execution to promptly grasp environmental trends and formulate corresponding response strategies.

Through sensitivity analysis, our team has gained a clearer understanding of the critical influencing factors and potential risk points in the spacecraft migration plan. This insight will facilitate more precise situational awareness, flexible strategy adjustments, and the formulation of effective risk mitigation measures during practical operations, ultimately ensuring the smooth execution and sustainable development of the migration plan.

7 Model Evaluation and Generalization

Advantages of the Model

In problem 1, due to the differences in age, physical activity intensity and other factors of different passenger groups, the material demands of individual passengers are different. At the same time, the load limitation of each spacecraft, the distribution ratio of passengers to materials and the overall goal of minimizing the number of spacecraft are comprehensively considered. Through iterative optimization, the optimal configuration of spacecraft is determined, which takes into account the complexity of material distribution and maximizes the carrying capacity of spacecraft, and provides scientific decision support for large-scale space migration.

In problem 2, there is a quantitative relationship between the actual resource acquisition and the ideal resource acquisition, and the difficulty coefficient can be used as a medium to clearly express the relationship between the two. Therefore, it is preferred to find a suitable functional relationship, which can not only describe the relationship between the two quantitatively, but also reflect the change of actual resource acquisition. By searching and comparing suitable functions, the exponential function is selected as the carrier. The nonlinear programming model of Monte Carlo simulation and dynamic programming realizes multi-modal decision processing, and successfully adjusts the coping strategy dynamically to adapt to complex risk situations.

Question 3: The fourth problem is to solve the integrated decision problem in dynamic environment. The main purpose is to simulate a decision-making process of resource acquisition and personnel transportation in order to try to achieve the delivery of a specified number of migrants to a safe area in a given time, and minimize costs and risks. Therefore, the proposed dynamic environment multi-objective optimization decision model comprehensively considers multiple objectives, and analyzes decisions through greedy strategy, including resource acquisition and personnel transportation decisions, and introduces the dynamic change of the environment, making each step consider the currently available information, which greatly improves the practical applicability of the model.

Disadvantages of the Model

Problem 1: Despite the combination of interpolation fitting and binary search iteration, in some cases, the scope of search may be too broad or not fine enough, resulting in a slight decline in algorithm performance, and the data may be single and lead to inaccurate results.

Problem 2: The quality of the results is highly dependent on the number and quality of the selected samples. If the initial samples are unevenly distributed or the number is too small, the opportunity to find the global optimal solution may be missed. Local search may fall into a local optimal solution and fail to find a global optimal. In some cases, the algorithm may loop over and over again for a certain amount of time without producing substantial improvements. In the process of Monte Carlo sampling and local search, the parameter setting of the algorithm (such as

sample size, search step size, etc.) has a great influence on the results, and improper selection may reduce the effectiveness of the solution.

Problem 3: Although the Monte Carlo simulation and dynamic programming model take into account the uncertainty of spacecraft failure, there may be other unknown or unpredictable factors that affect the success rate and cost of the mission, and the impact factor of the model is relatively simple and the implementation is low.

Problem 4: When greedy strategy is adopted in the multi-objective optimization decision model for dynamic environment, the changes of resource acquisition efficiency, difficulty coefficient and failure probability are simplified into linear functions, which are usually nonlinear changes, which may not be enough to reflect the complexity of real environment.

Model Promotion

Problem 1 model can be applied to other transportation scenarios, such as logistics transportation of different volumes of goods in the same vehicle, to simulate personnel evacuation and resource allocation in natural disasters or other emergency situations, such as rescue operation planning after earthquakes, floods or fires.

The dynamic environment multi-objective optimization decision model in question 4 can be used to simulate resource management and sustainable development plan decision support in resource-limited and environmentally fragile environments, such as forest resource management or Marine resource protection.

8 References

- [1]Tao Wang,and Zhiping Chen."Optimal portfolio and insurance strategy with biometric risks, habit formation and smooth ambiguity."Insurance Mathematics and Economics 118.(2024):
- [2]Khamiss Cheikh, et al."Balancing the maintenance strategies to making decisions using Monte Carlo method."MethodsX 13.(2024):
- [3]Harmeet Kaur,and Shubham Gargish."DRAP-CPU: a novel vm migration approach through a dynamic prioritized resource allocation strategy."Microsystem Technologies prepubulish(2024):
- [4]Agustín Gabriel Yabo,Jean Baptiste Caillau,and Jean Luc Gouzé."Optimal Bacterial Resource Allocation Strategies in Batch Processing."SIAM Journal on Applied Mathematics (2024):
- [5]Javier Urrios,Guillermo Pacheco Ramos,and Rafael Vazquez."Optimal planning and guidance for Solar System exploration using Electric Solar Wind Sails."Acta Astronautica 217.(2024):
- [6]JiaZhu, et al."Optimal resource allocation strategy for integrated sensing and communication cellular network with multiple user demands."Electronics Letters 60.4(2024):
- [7]Ziqing Zhang."Multi-regime foreign exchange rate model: Calibration and pricing."Mathematics and Computers in Simulation 220.(2024):
- [8]Shahcheraghi Seyed Hadi,Dianatpour Mehdi,and Hayati Mohammad."Modelling and estimating the Blaine number of iron ore concentrate by response surface methodology and Monte Carlo simulation."Advanced Powder Technology 34.12(2023):
- [9]Muthirevula Sharmila,and Ravinutala Venkata Subbu Satyanarayana."An intelligent resource allocation strategy for machine type communication environment."International Journal of Communication Systems 37.1(2023):
- [10]Suresh Chintalapudi V., et al."Optimal renewable resource for enhanced performance of unbalanced radial distribution systems."Microsystem Technologies 29.4(2023):